

Research Article

On the Probabilistic Proof of the Convergence of the Collatz Conjecture

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A new approach towards probabilistic proof of the convergence of the Collatz conjecture is described via identifying a sequential correlation of even natural numbers by divisions by 2 that follows a recurrent pattern of the form $x, 1, x, 1 \dots$, where x represents divisions by 2 more than once. The sequence presents a probability of 50:50 of division by 2 more than once as opposed to division by 2 once over the even natural numbers. The sequence also gives the same 50:50 probability of consecutive Collatz even elements when counted for division by 2 more than once as opposed to division by 2 once and a ratio of 3:1. Considering Collatz function producing random numbers and over sufficient number of iterations, this probability distribution produces numbers in descending order that lead to the convergence of the Collatz function to 1, assuming that the only cycle of the function is 1-4-2-1.

1. Introduction

The Collatz conjecture concerns natural numbers treated as $(\text{mod } 2)$ of positive even integers. It is defined by the function

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \equiv 0 \pmod{2} \\ 3n + 1 & \text{if } n \equiv 1 \pmod{2} \end{cases} \quad (1)$$

It simply asks you to keep dividing any positive even integer repeatedly by 2 until it becomes an odd integer, then convert it to even integer by tripling it and adding 1 to it, and then repeat the process. The conjecture has been widely studied [1, 2]. It predicts that the recurring process will always form a sequence that descends on the natural numbers to cycle around the trivial cycle 1-4-2-1. The conjecture involves the natural numbers and it simply asks, under any complete process of the conjecture, why it is always the case that, over statistically sufficient number of iterations, the decrease made by the divisions by 2 exceeds the increase made by the conversions from oddness to evenness. It has been noticed here that, from a start odd positive integer, one iteration either increases the number when the result-even number is divided by 2 *only once* to obtain an odd number or decreases the start number when the result-even number

is divided by 2 *more than once*. Therefore, we seek here to quantify the decrease and the increase probabilistically of the start number after every iteration and generalize that over a sufficient number of iterations to check convergence of the function. It is claimed here that the function decreases the start number until it reaches a cycle, because statistically the sequence of all of the consecutive even integers of the elements of the Collatz function over the natural numbers (validated by deducting 1 from every even positive integer and then checking divisibility by 3) has a recurrent pattern of $x, 1, x, 1 \dots$ of division by 2 more than once compared to division by 2 once of probability 50:50 and ratio of about 3:1, where x is division by 2 more than once.

Collatz conjecture function seems to produce random numbers and generate a random walk process locally but globally converges to 1. Therefore, to prove the convergence of the conjecture probabilistically it is sufficient to show that globally the recurrence of divisions of Collatz even elements by 2 more than once to reach an odd number has the same probability as that of their recurrent divisions by 2 once, denoted here as recurrent frequency (RF), and averages by the ratio of about 3:1. Summing over the respective divisions will always lead by a margin that offsets the increase of the recurrent sum made by the recursive conversion process of

the odd Collatz number to even number by tripling it and adding 1 to it. This is easily noticeable if we recognize that if the positive even integers were sequenced by increase by 2, e.g., 2, 4, 6, 8, ..., division by 2 over the positive even integers follows a sequential order that is described as follows: if any of the sequence's even elements produces an odd number when divided by 2 once, the following element in the sequence must produce an odd number by division by 2 more than once. This hidden regularity produces a 50-50 probabilistic RF of division by 2 over the positive even integers and turns what seems a random distribution of division by 2 to a global process that makes the events of division by 2 recurrence over the whole positive even integers progress according to the sequence $x, 1, x, 1 \dots$, where x is the number of divisions of the even number by 2 more than once to produce an odd number. Here we prove that Collatz-even numbers also follow the same 50:50 probability distribution that leads to descent convergence of the sequence made by the function to a cycle. The proposed proof of the Collatz conjecture here is complete if its process only cycles about 1, 4, and 2, since the decrease of the sequence of the global Collatz process is assembled from perfect correlated probabilistic events defined by the sequence $x, 1, x, 1 \dots$, over the function's even elements. This probabilistic correlation is not heuristically derived as opposed to the well-known heuristic argument of the function found in many references [3–5] which states that the function averages division by 2 once $1/2$ of the time and division by 2 twice $1/4$ of the time and division by 2 three times $1/8$ of the time, etc., which produces a decrease of $3/4$ of the preceding number each iteration on average. In this paper it is claimed that the function $3n + 1$ produces an increase of the odd start number of 50%, $1/2$ of the time as opposed to a decrease of the odd start number of 62% the other $1/2$ of the time, averaged over a sample of sufficiently large number of Collatz even integers if we assume that the mixing properties of the function's even integers are truly picked at random in the process.

2. Division by 2 Sequence of Positive Even Integers

For comparison and to easily identify the RF sequence of division by 2 for Collatz function elements, we first generate the RF sequence of positive even integers.

Lemma 1. *Let n be any positive even integer that can be divided by 2 only once to yield an odd positive integer; then the next even integer $n + 2$ must be divided by 2 by more than once to yield an odd positive integer.*

Proof. If $n/2 \rightarrow \text{odd}$ by initial definition, then $(n + 2)/2 \rightarrow \text{parity}$. Adding the LHS expressions yields $n + 1$, an odd number. This necessitates that $(n + 2)/2 \rightarrow \text{even}$ and the term $n + 2$ is divisible by 2 more than once. $\square$

Lemma 2. *Let n be any positive even integer that can be divided by 2 only once to yield an odd positive integer; then the second to next even integer $n + 4$ must be divided by 2 only once to yield an odd positive integer.*

Proof. If $n/2 \rightarrow \text{odd}$ by initial definition, then $(n + 4)/2 \rightarrow \text{parity}$. Adding the LHS expressions yields $n + 2$, an even number. This necessitates $(n + 4)/2 \rightarrow \text{odd}$ and the term $n + 4$ is divisible by 2 only once to obtain an odd number. $\square$

From Lemmas 1 and 2, we generate a table of positive even integers and their corresponding frequencies of division by 2 until reaching an odd parity. Starting with the first row as the even integers made by the term 2^s , $s \in \mathbb{Z}^+$ with elements as the frequencies of division by 2, and spanning the natural numbers we can construct a “RF table” over all positive integers that identifies Collatz elements with the back-bone as the line of integers that collapse to 1 by repeated divisions by 2 made by the even numbers 2^s , as Collatz function requires. This row makes a symmetrical line that contains all even numbers made by Collatz function that collapse to the trivial cycle 1-4-2-1, e.g., 4, 8, 16, 32. We then construct columns in ascending order by increase by 2 to produce all even positive integers with each column ending by an even number that is two less than the next integer on the collapsing symmetrical line. We observe that the symmetrical line in the table has symmetrical sequential frequencies for all of the columns to infinity along the rows and makes rows with equal frequencies because of the ordered repeated frequencies for each column, which allows us to estimate relative RFs, a key probability distribution that allows us to conclude that Collatz conjecture converges probabilistically to a cycle. The table is constructed in this order mainly to be able to count frequencies of divisions by 2 and approximate the relative RFs of even positive integers to yield an odd number. It follows that consecutive Collatz function's even elements (in italic) also follow the same pattern as those of the sequence of the table of $x, 1, x, 1 \dots$. We also construct the table with the variable s spanning all positive integers of 2^s on the symmetrical line, not just even Collatz function elements that contribute to the collapse process of Collatz function, to produce a line of all powers of twos.

Starting with any natural number, Collatz function produces numbers in seemingly random way locally but globally the numbers decrease and the process proceeds toward the collapsing symmetrical line and to the left on the table and it eventually hits a number on the symmetrical line and then collapses to 1 and cycles around 1-4-2-1 in a deterministic process.

The symmetrical distribution of frequencies of divisions by 2 of even natural numbers as in the table exhibits a classical probability distribution about the collapsing symmetrical line over the natural numbers. Only those numbers on the symmetrical line that satisfy Collatz function can branch out and contribute to the collapse process to 1 (those numbers with s an even integer) and the branches that are connected via the function with odd start numbers making new subbranches on the Collatz tree (see Figure 1) that if a branch is reached, the process will collapse to its start odd number; i.e., on the trunk of the tree, the number 2^8 (256) contributes to the collapse process because you can deduct 1 from it and divide by 3 to get a whole number, but the number 2^9 (512) does not, and the number 341 leads to 2^{10} (1024) on the symmetrical line that collapses to 1 while the odd number 357913941 ends with 2^{30}

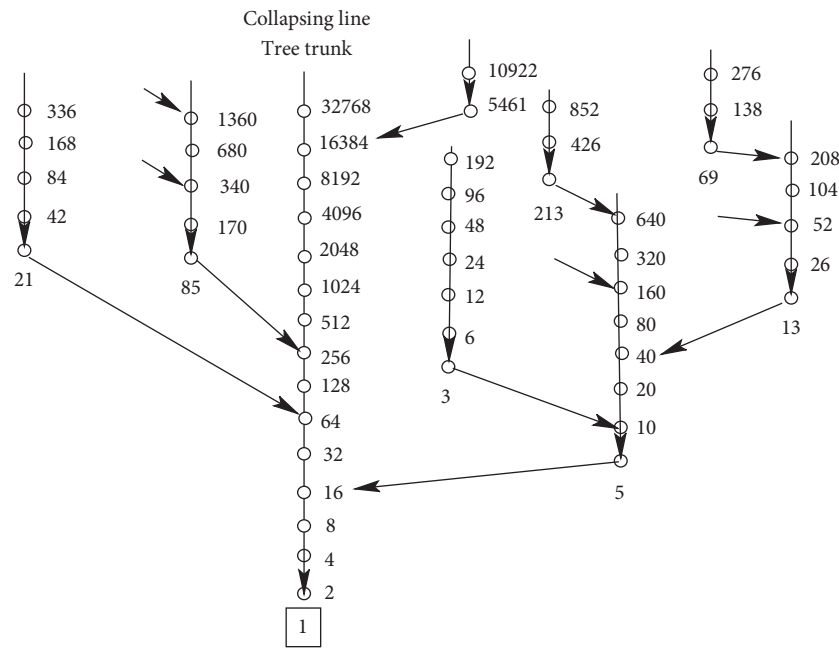


FIGURE 1: Representative of Collatz tree.

(1073741824) on the symmetrical line as well. Those numbers on the symmetrical line that can be traced backward by the function $3n + 1$ act as points for branching out to trace the Collatz tree where the symmetrical line is the tree's trunk.

3. Perfect Symmetry of the Table of Positive Even Integers with Pivotal Frequencies

Looking for hidden symmetry in the background of even integers in terms of RFs is of prime importance to assign symmetry to the RFs of the $3n + 1$ function and determine their probability ratios. That is because Collatz function's elements occur sequentially every third element in the table of positive integers as represented by Table 2. Quick observation of the table of positive even integers reveals that each column is exactly symmetrical about a pivotal new frequency that is next number on the number line to the pivotal frequency in the preceding column and equals $(s - 1)$ of its own; i.e., the column 2^5 (32) has double the frequencies of the preceding column of 2^4 about the pivotal RF of 4, with pivotal RF greater than the preceding one of 3 of the preceding column by 1 and equal to $(S - 1)$, where s is 5. Those pivotal RFs increase to infinity by the increase of the numbers of the column about the symmetrical line of 2^5 .

4. Perfect Symmetry of the Table of $3n + 1$ Even Elements

The perfect symmetrical RFs of division by 2 of the even terms of the $3n + 1$ function allow us to determine their probability ratios. We construct Table 3 the same way we

construct Table 1 by building columns in ascending order of consecutive elements about the collapsing symmetrical line that is made of $3n + 1$ elements. Similar to the columns of the even integers in Table 1, the columns of even elements of $3n + 1$ of Table 3 are symmetrically distributed since each row in the table carries the same frequency for all of the columns and each column begins with the same RF sequence as the preceding column. This symmetry is important since it spans positive integers to infinity and it allows taking the average of the RFs of the elements of the function $3n + 1$ that divide by 2 more than once as a measure of the behavior of the function's complete process that leads to a descending trajectory. The ratio of about 3:1 of divisions by 2 more than once of the RFs compared to divisions by 2 once was computed from the table with counts varying by length and location, between 100 and 3000 of consecutive $3n + 1$ elements and up to 2^{36} element. While the RF ratio of about 3:1 seems to be consistent to infinity as shown by Table 3, this work needs yet elaborative approach to identify a proper distribution that yields this ratio to prove the consistency to infinity of the RF symmetry of Table 3 to reach full proof of the convergence of Collatz conjecture. Disregarding the exact shape of the proper distribution, here we use symmetry of the distribution of the $3n + 1$ even elements about the symmetrical line to estimate the RF ratio of 3:1.

5. Probability Distribution of Even Natural Numbers in Terms of Division by 2

The ordered distribution of even natural numbers in terms of division by 2 about the symmetrical line represents a classical probability distribution.

TABLE 1: Positive even integers and their corresponding frequencies of division by 2. Any row has the same frequency. Even elements of $3n + 1$ function are in italic.

#	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	
2 ^s	3	8	4	16	5	32	6	64	7	128	8	256	9	512	10	1024	30	1073741824	
2	1	10	1	18	1	34	1	66	1	130	1	258	1	514	1	1026	1	1073741826	
3	2	12	2	20	2	36	2	68	2	132	2	260	2	516	2	1028	2	1073741828	
4	1	14	1	22	1	38	1	70	1	134	1	262	1	518	1	1030	1	1073741830	
5			3	24	3	40	3	72	3	136	3	264	3	520	3	1032	3	1073741832	
6			1	26	1	42	1	74	1	138	1	266	1	522	1	1034	1	1073741834	
7			2	28	2	44	2	76	2	140	2	268	2	524	2	1036	2	1073741836	
8			1	30	1	46	1	78	1	142	1	270	1	526	1	1038	1	1073741838	
9				4	48	4	80	4	144	4	272	4	528	4	1040	4	1073741840		
10				1	50	1	82	1	146	1	274	1	530	1	1042	1	1073741842		
11				2	52	2	84	2	148	2	276	2	532	2	1044	2	1073741844		
12				1	54	1	86	1	150	1	278	1	534	1	1046	1	1073741846		
13				3	56	3	88	3	152	3	280	3	536	3	1048	3	1073741848		
14				1	58	1	90	1	154	1	282	1	538	1	1050	1	1073741850		
15				2	60	2	92	2	156	2	284	2	540	2	1052	2	1073741852		
16				1	62	1	94	1	158	1	286	1	542	1	1054	1	1073741854		
17					5	96	5	160	5	288	5	544	5	1056	5	1073741856			
18					1	98	1	162	1	290	1	546	1	1058	1	1073741858			
19					2	100	2	164	2	292	2	548	2	1060	2	1073741860			
20					1	102	1	166	1	294	1	550	1	1062	1	1073741862			
21					3	104	3	168	3	296	3	552	3	1064	3	1073741864			
22					1	106	1	170	1	298	1	554	1	1066	1	1073741866			
23					2	108	2	172	2	300	2	556	2	1068	2	1073741868			
24					1	110	1	174	1	302	1	558	1	1070	1	1073741870			
25					4	112	4	176	4	304	4	560	4	1072	4	1073741872			
26					1	114	1	178	1	306	1	562	1	1074	1	1073741874			
27					2	116	2	180	2	308	2	564	2	1076	2	1073741876			
28					1	118	1	182	1	310	1	566	1	1078	1	1073741878			
29					3	120	3	184	3	312	3	568	3	1080	3	1073741880			
30					1	122	1	186	1	314	1	570	1	1082	1	1073741882			
31					2	124	2	188	2	316	2	572	2	1084	2	1073741884			
32					1	126	1	190	1	318	1	574	1	1086	1	1073741886			
33							6	192	6	320	6	576	6	1088	6	1073741888			
34								1	194	1	322	1	578	1	1090	1	1073741890		
35								2	196	2	324	2	580	2	1092	2	1073741892		
36								1	198	1	326	1	582	1	1094	1	1073741894		
37								3	200	3	328	3	584	3	1096	3	1073741896		
38								1	202	1	330	1	586	1	1098	1	1073741898		
39								2	204	2	332	2	588	2	1100	2	1073741900		
40								1	206	1	334	1	590	1	1102	1	1073741902		
41								4	208	4	336	4	592	4	1104	4	1073741904		
42								1	210	1	338	1	594	1	1106	1	1073741906		
43								2	212	2	340	2	596	2	1108	2	1073741908		
44								1	214	1	342	1	598	1	1110	1	1073741910		
45								3	216	3	344	3	600	3	1112	3	1073741912		
46								1	218	1	346	1	602	1	1114	1	1073741914		
47								2	220	2	348	2	604	2	1116	2	1073741916		
48								1	222	1	350	1	606	1	1118	1	1073741918		
49								5	224	5	352	5	608	5	1120	5	1073741920		

TABLE 1: Continued.

#	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.	F	Int.
50									1	226	1	354	1	610	1	1122	1	1073741922
51									2	228	2	356	2	612	2	1124	2	1073741924
52									1	230	1	358	1	614	1	1126	1	1073741926
53									3	232	3	360	3	616	3	1128	3	1073741928
54									1	234	1	362	1	618	1	1130	1	1073741930
55									2	236	2	364	2	620	2	1132	2	1073741932
56									1	238	1	366	1	622	1	1134	1	1073741934
57									4	240	4	368	4	624	4	1136	4	1073741936
58									1	242	1	370	1	626	1	1138	1	1073741938
59									2	244	2	372	2	628	2	1140	2	1073741940
60									1	246	1	374	1	630	1	1142	1	1073741942
61									3	248	3	376	3	632	3	1144	3	1073741944
62									1	250	1	378	1	634	1	1146	1	1073741946
63									2	252	2	380	2	636	2	1148	2	1073741948
64									1	254	1	382	1	638	1	1150	1	1073741950
65											7	384	7	640	7	1152	7	1073741952
									..	...	...	...	...	...	...	...	...	...

TABLE 2: Representative of pivotal RF in the table of positive even integers.

#	F	Int.	F	Int.
2^s	4	16	5	32
1	1	18	1	34
2	2	20	2	36
3	1	22	1	38
4	3	24	3	40
5	1	26	1	42
6	2	28	2	44
7	1	30	1	46
8			4	48
9			1	50
10			2	52
11			1	54
12			3	56
13			1	58
14			2	60
15			1	62

Lemma 3. *The probability of division by 2 more than once and division by 2 once for a randomly chosen positive even number is P (more than once) = P (once) = $1/2$.*

Proof. It follows from the ordered distribution of division by 2 once followed by division by 2 more than once by Lemma 1 and Lemma 2. The probability is easy to check in Table 1. $\square$

6. Probability Distribution of Collatz Function's Even Elements

Table 3 represents 50:50 probability of the Collatz function's even elements in terms of their RFs of division by 2 more than once as opposed to division by 2 once.

Lemma 4. *Collatz function even elements are sequenced every three consecutive numbers on the sequence of the even nonnegative integers with probability of division by 2 more than once as opposed to division by 2 once, to obtain an odd number for a randomly chosen Collatz element, being P (more than once) = P (once) = $1/2$.*

Proof. Let n be any Collatz even element that can be divided by 3 after subtracting 1; then the next even integer $n+2$ is not a Collatz even element since it does not follow that restriction, neither is the next one, but the one that follows is, since a Collatz element is restricted by

$$(n+k) - 1 \quad (2)$$

which is divisible by 3 if the variable k is multiples of 6 only and leads to the fourth consecutive integer after the variable n on the table of nonnegative even integers.

Further, let n be any Collatz even element that is only divisible by 2 once; the following set of logical equations describes RFs of all of Collatz even elements by obtaining their parity.

First, if $n/2 \rightarrow \text{odd}$ by initial definition, then $(n+6)/2 \rightarrow \text{parity}$. Adding the LHS expressions yields $n+3$, an odd number. This necessitates that $(n+6)/2 \rightarrow \text{even}$ and the term $n+6$ is even number and therefore it is divisible by 2 more than once to obtain an odd number.

TABLE 3: Representative of symmetrical RFs in the table of $3n + 1$ even elements.

#	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$
2^5	4	16	6	64	8	256	10	1024	12	4096	24	16777216
1	1	22	1	70	1	262	1	1030	1	4102	1	16777222
2	2	28	2	76	2	268	2	1036	2	4108	2	16777228
3	1	34	1	82	1	274	1	1042	1	4114	1	16777234
4	3	40	3	88	3	280	3	1048	3	4120	3	16777240
5	1	46	1	94	1	286	1	1054	1	4126	1	16777246
6	2	52	2	100	2	292	2	1060	2	4132	2	16777252
7	1	58	1	106	1	298	1	1066	1	4138	1	16777258
8			4	112	4	304	4	1072	4	4144	4	16777264
9			1	118	1	310	1	1078	1	4150	1	16777270
10			2	124	2	316	2	1084	2	4156	2	16777276
11			1	130	1	322	1	1090	1	4162	1	16777282
12			3	136	3	328	3	1096	3	4168	3	16777288
13			1	142	1	334	1	1102	1	4174	1	16777294
14			2	148	2	340	2	1108	2	4180	2	16777300
15			1	154	1	346	1	1114	1	4186	1	16777306
16			5	160	5	352	5	1120	5	4192	5	16777312
17			1	166	1	358	1	1126	1	4198	1	16777318
18			2	172	2	364	2	1132	2	4204	2	16777324
19			1	178	1	370	1	1138	1	4210	1	16777330
20			3	184	3	376	3	1144	3	4216	3	16777336
21			1	190	1	382	1	1150	1	4222	1	16777342
22			2	196	2	388	2	1156	2	4228	2	16777348
23			1	202	1	394	1	1162	1	4234	1	16777354
24			4	208	4	400	4	1168	4	4240	4	16777360
25			1	214	1	406	1	1174	1	4246	1	16777366
26			2	220	2	412	2	1180	2	4252	2	16777372
27			1	226	1	418	1	1186	1	4258	1	16777378
28			3	232	3	424	3	1192	3	4264	3	16777384
29			1	238	1	430	1	1198	1	4270	1	16777390
30			2	244	2	436	2	1204	2	4276	2	16777396
31			1	250	1	442	1	1210	1	4282	1	16777402
32					6	448	6	1216	6	4288	6	16777408
33					1	454	1	1222	1	4294	1	16777414
34					2	460	2	1228	2	4300	2	16777420
35					1	466	1	1234	1	4306	1	16777426
36					3	472	3	1240	3	4312	3	16777432
37					1	478	1	1246	1	4318	1	16777438
38					2	484	2	1252	2	4324	2	16777444
39					1	490	1	1258	1	4330	1	16777450
40					4	496	4	1264	4	4336	4	16777456
41					1	502	1	1270	1	4342	1	16777462
42					2	508	2	1276	2	4348	2	16777468
43					1	514	1	1282	1	4354	1	16777474
44					3	520	3	1288	3	4360	3	16777480
45					1	526	1	1294	1	4366	1	16777486
46					2	532	2	1300	2	4372	2	16777492
47					1	538	1	1306	1	4378	1	16777498
48					5	544	5	1312	5	4384	5	16777504

TABLE 3: Continued.

#	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$
49					1	550	1	1318	1	4390	1	16777510
50					2	556	2	1324	2	4396	2	16777516
51					1	562	1	1330	1	4402	1	16777522
52					3	568	3	1336	3	4408	3	16777528
53					1	574	1	1342	1	4414	1	16777534
54					2	580	2	1348	2	4420	2	16777540
55					1	586	1	1354	1	4426	1	16777546
56					4	592	4	1360	4	4432	4	16777552
57					1	598	1	1366	1	4438	1	16777558
58					2	604	2	1372	2	4444	2	16777564
59					1	610	1	1378	1	4450	1	16777570
60					3	616	3	1384	3	4456	3	16777576
61					1	622	1	1390	1	4462	1	16777582
62					2	628	2	1396	2	4468	2	16777588
63					1	634	1	1402	1	4474	1	16777594
64					7	640	7	1408	7	4480	7	16777600
65					1	646	1	1414	1	4486	1	16777606
66					2	652	2	1420	2	4492	2	16777612
67					1	658	1	1426	1	4498	1	16777618
68					3	664	3	1432	3	4504	3	16777624
69					1	670	1	1438	1	4510	1	16777630
70					2	676	2	1444	2	4516	2	16777636
71					1	682	1	1450	1	4522	1	16777642
72					4	688	4	1456	4	4528	4	16777648
73					1	694	1	1462	1	4534	1	16777654
74					2	700	2	1468	2	4540	2	16777660
75					1	706	1	1474	1	4546	1	16777666
76					3	712	3	1480	3	4552	3	16777672
77					1	718	1	1486	1	4558	1	16777678
78					2	724	2	1492	2	4564	2	16777684
79					1	730	1	1498	1	4570	1	16777690
80					5	736	5	1504	5	4576	5	16777696
81					1	742	1	1510	1	4582	1	16777702
82					2	748	2	1516	2	4588	2	16777708
83					1	754	1	1522	1	4594	1	16777714
84					3	760	3	1528	3	4600	3	16777720
85					1	766	1	1534	1	4606	1	16777726
86					2	772	2	1540	2	4612	2	16777732
87					1	778	1	1546	1	4618	1	16777738
88					4	784	4	1552	4	4624	4	16777744
89					1	790	1	1558	1	4630	1	16777750
90					2	796	2	1564	2	4636	2	16777756
91					1	802	1	1570	1	4642	1	16777762
92					3	808	3	1576	3	4648	3	16777768
93					1	814	1	1582	1	4654	1	16777774
94					2	820	2	1588	2	4660	2	16777780
95					1	826	1	1594	1	4666	1	16777786
96					6	832	6	1600	6	4672	6	16777792
97					1	838	1	1606	1	4678	1	16777798
98					2	844	2	1612	2	4684	2	16777804

TABLE 3: Continued.

#	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$	F	$3n + 1$
99					1	850	1	1618	1	4690	1	16777810
100					3	856	3	1624	3	4696	3	16777816
101					1	862	1	1630	1	4702	1	16777822
102					2	868	2	1636	2	4708	2	16777828
103					1	874	1	1642	1	4714	1	16777834
104					4	880	4	1648	4	4720	4	16777840
105					1	886	1	1654	1	4726	1	16777846
106					2	892	2	1660	2	4732	2	16777852
107					1	898	1	1666	1	4738	1	16777858
108					3	904	3	1672	3	4744	3	16777864
109					1	910	1	1678	1	4750	1	16777870
110					2	916	2	1684	2	4756	2	16777876
111					1	922	1	1690	1	4762	1	16777882
112					5	928	5	1696	5	4768	5	16777888
113					1	934	1	1702	1	4774	1	16777894
114					2	940	2	1708	2	4780	2	16777900
115					1	946	1	1714	1	4786	1	16777906
116					3	952	3	1720	3	4792	3	16777912
117					1	958	1	1726	1	4798	1	16777918
118					2	964	2	1732	2	4804	2	16777924
119					1	970	1	1738	1	4810	1	16777930
120					4	976	4	1744	4	4816	4	16777936
121					1	982	1	1750	1	4822	1	16777942
122					2	988	2	1756	2	4828	2	16777948
123					1	994	1	1762	1	4834	1	16777954
124					3	1000	3	1768	3	4840	3	16777960
125					1	1006	1	1774	1	4846	1	16777966
126					2	1012	2	1780	2	4852	2	16777972
127					1	1018	1	1786	1	4858	1	16777978
128							8	1792	8	4864	8	16777984
							...	...	...	...	...	...

Second, if $n/2 \rightarrow \text{odd}$ by initial definition, then $(n + 12)/2 \rightarrow \text{parity}$. Adding the LHS expressions yields $n + 6$, an even number. This necessitates that $(n + 12)/2 \rightarrow \text{odd}$ and that the term $n + 12$ is divisible by 2 only once to obtain an odd number. $\square$

This is shown by quick inspection of the sequence of the even natural numbers by subtracting 1 followed by division by 3 (italic face in Table 1).

Note. Lemma 4 can be generalized to any generalized Collatz function in the form of $kn + 1$, where k is an odd number and their corresponding sequence on the even nonnegative integers sequence can be derived accordingly.

7. Probability Ratios of RFs of Collatz Even Elements

Since RFs are ordered perfectly among all Collatz even elements as Table 3 indicates, any sufficiently large sample is a

true representation to compute RF ratio of Collatz function's elements.

Lemma 5. *The sum of divisions by 2 more than once is on average 2.97 times (about 3 times) the sum of divisions by 2 once over the Collatz even elements over the first 1500 counts.*

Proof. Inspection of Table 3 verifies the prediction. $\square$

Theory 1. Collatz function process must produce a descending order of numbers over adequate number of iterations with 3:1 RF ratio of division by 2 of even $3n + 1$ elements.

Proof. Considering the apparent random distribution of even integers produced by Collatz function processes, the probability of division by 2 more than once to division by 2 once of Collatz elements is $1/2$. Therefore we can average Collatz processes as two distinctive operations of a start number of an odd Collatz element that ends up in an odd number. The first operation is to increment it by tripling it and adding 1

TABLE 4: Selective RF ratios of the Collatz even elements samples.

Sample space	Ratio
1-100	2.88:1
101-200	2.96:1
201-300	2.92:1
301-400	2.94:1
401-500	3.04:1
1-500	2.92:1
200-1000	2.97:1
1-1460	2.96:1

and then dividing by 2 once, which increases the start number while the other operation is to increment the start number but divide it by 2 three times, which decreases the start number by a larger magnitude than the increase, in line with Collatz conjecture. To see this, let n be the start odd-number. Then applying the function $3n + 1$ and dividing by 2 once give an end number,

$$End = \frac{3n + 1}{2} \quad (3)$$

$$End = 1.5n + 0.5$$

For large enough numbers, this equation gives an increase of about 1/2 of the start number. If we divide the function by 2^3 instead, the process gives

$$End = \frac{3n + 1}{2^3} \quad (4)$$

$$End = 0.375n + 0.125$$

This equation gives a decrease of about 3/5 of the start number, which is larger than the increase, and leads to successive average decrease of the start number. $\square$

The *critical ratio* that produces reduction of the start number as increase is about 2.57:1.

Collatz function then produces steps that end with even numbers that zigzag up and down but in a descending manner until it eventually reaches an odd number whose ascending step is on the symmetrical line and collapses to the ultimate odd number of 1 assuming the cycle 1-4-2-1 is the only cycle in the process.

For low counts of $3n + 1$ even elements, Table 4 shows selective ratios of the RFs of division by 2 more than once compared to division by 2 once. The ratios reveal that, for low counts, as low as 100 elements that may span a Collatz process, the process exhibits decreasing trajectories. Also, the ratio of the first 129 RFs of the column under 2^{24} on the symmetrical line in Table 3 is found to be 2.92, which yields a decreasing trajectory since all of the ratios are above the critical ratio.

For very low counts of a complete process, the process may quickly end its life and collapse to 1 by hitting the symmetrical line with increasing trajectory up to the symmetrical line; examples are start odd numbers of 3 and 5.

Example 6 (start odd number is 3). For a 50:50 probability of division by 2 more than once as opposed to division by 2 once with ratio 3:1 and a repeating pattern, the iteration process of applying the function $3n + 1$ gives 10. Division by 2 once gives 5. If we divide by 2 three times the iteration gives 1.25. This is actually an increase of the start number. Fortunately, the process hits the symmetrical line and collapses to 1 upon increase of the start number only upon repeating the process with the number 5. Starting with the number 7 and up on the number line, in general, the collective decrease of the start number is larger than the collective increase, except for those start numbers that hit the symmetrical line before the decrease occurs, e.g., start numbers that make the main branches on the Collatz tree (see Figure 1) such as 5, 21, 85, 5461, etc., in line with theory 1.

Example 7 (start odd number is 9999). Incrementing it by Collatz function $3n+1$ and dividing by 2 once yield 14999. This is an increase of the number of 50%. Repeating the process but dividing by 2 three times, you get 3749.75. That is a decrease of the start number of about 62%. Obviously, the percentage decrease is larger than the percentage increase of the start number in line with the Collatz conjecture.

8. Comparison with Generalized Collatz Functions

Many generalized Collatz functions are discussed in the literature [4–8]. Generalized Collatz functions such as $5n + 1$ and $7n + 1$ do have probability distributions of their even elements in terms of division by 2; i.e., inspection of Table 1 reveals that the function $5n + 1$ has even integers every four consecutive integers over the even integers sequence with a 50:50 ratio of division by 2 more than once as opposed to once. The same goes with the function $7n + 1$ with spacing of seven consecutive integers. Therefore, to check for the divergence of those functions, we must compute the relative frequencies of divisions by 2 that contribute to the rise of the start number as opposed to those that contribute to its descending. It is noticed here that, unlike the function $3n + 1$, division by 2 contributes differently to the increase or decrease of the start number for other generalized functions; i.e., besides the fact that you multiply the start number by 5 instead of 3, division by 2 once as well as twice with the function $5n + 1$ contributes to the rise of the start number and its process and then produces an equation with coefficient of n much large compared to the function $3n + 1$ leading to the divergence of the function.

Question. Why does not the function $5n + 1$ eventually reach the symmetrical line on its ascending trajectory and it collapses to 1?

Answer. None of the even numbers on the symmetrical line belongs to the function's elements since deducting 1 from all of its even integers does not produce odd integers that are evaluated to 0 (mod 5).

9. The Only Trivial Cycle of the Collatz Function Is 1-4-2-1

It is easy to prove that the cycle 1-4-2-1 is the only trivial cycle for Collatz function.

Lemma 8. *Let n be any positive odd integer. Then the only trivial cycle of the Collatz function is 1-4-2-1.*

Proof. The equation

$$n = \frac{3n + 1}{2^l} \quad (5)$$

describes a trivial cycle where l is an integer that equals the number of divisions by 2. Solving for n yields

$$n = \frac{1}{2^l - 3} \quad (6)$$

For positive integer solution, l must be 2, n must be 1, and $3n + 1$ must be 4. That is because if $l > 2$, the expression yields n as fraction and if $l < 2$, the expression yields negative value. This leaves $l = 2$ the only solution to the equation with a start number 1. $\square$

10. Nontrivial Nested Cycles

Collatz conjecture forbids looping anywhere on the Collatz tree except at the bottom of the trunk as clear in Figure 1. Starting at any point on the tree, Collatz function allows the process only to head in one direction from one point to another on a subbranch to another subbranch leading to a main branch and then to the trunk and finally collapsing to the loop 1-4-2-1. A global nested trajectory of Collatz conjecture is represented by the sequence that defines the trajectory of a start odd number n ,

$$n, T(n), T^{(2)}(n), T^{(3)}(n), \dots \quad (7)$$

The sequence $T^{(k)}(n)$ of the function must become periodic with an end number that equals the start number n for any $k \geq 1$ and $n \neq 1$.

Collatz conjecture suggests the nonexistence of a nested cycle with the start number that equals the end number. This may not be generalized locally for any relatively small degree of nesting of the function which, according to the conjecture, prohibits the return to the same start number different than 1. Since we assume that the function heads to stochastic behavior very fast, we may assume with small degree of certainty that the function does not trace back to the same start number and hit a cycle somewhere on the sequence of integers with high probability by the same reasoning of stochastic distribution of large number of elements in a sample, to hit the same number twice with a very low probability.

In comparison with the function $5n + 1$ that has no elements on the trunk of the Collatz tree and therefore any of its cycles must end up with an odd number other than 1

(see Figure 1), the function $3n + 1$ has elements on the trunk and it collapses to 1 if it happens that the function's trajectory reaches the trunk to cycle around the trivial cycle 1-4-2-1. Two known cycles of $5n + 1$ are 17-27-43-17 and 13-83-33-13. The higher degree of zigzagging up and down of the function (compared with the function $3n + 1$) starts with the column with a start odd number 13 of Collatz tree of the specified function, then alternates between other columns, here 81 and 33 for the second cycle, and returns to the column of start number 13 (see Figure 1). Any number of the four numbers can be the starting and ending number as well. Obviously, the three columns involved must contain elements of the function that are spaced wide enough for the trajectory to return to the same column (tree branch) it starts from. The generalized function $5n + 1$ statistically has a larger chance to hit the start branch after its launch than $3n + 1$ because the degree of zigzagging about its launch branch is higher because it has a wider spacing (multiplying by 5) and also that division by 2 once as well as twice contributes to the increase of the function as opposed to only division by 2 once that contributes to the increase of the start number for the function $3n + 1$. Overall, probabilistically, it does not seem there exists what prohibits a nontrivial cycle for the function $3n + 1$.

11. Conclusion

The convergence process of the $3n + 1$ function was proven over its elements up to 2^{36} by identifying a sequence of the function's positive even integers that produces a probability of 50:50 of the division of the integers by 2 more than once as opposed to their division by 2 once with a ratio of about 3:1. For any positive odd integer, the collective divisions by 2 more than once that produced a total decrease of the start number in the function's trajectory were found to exceed the total increase of the start number produced by division by 2 once. The process indicates a systematic global decrease until one event matches an even number on the symmetrical line and collapses to 1 and loops the cycle 1-4-2-1, presuming that the function yields no other cycles.

Data Availability

The data used to support the findings of this study are mostly available in the text. Any further data might be requested from the corresponding author.

Conflicts of Interest

The author declares that he has no conflicts of interest.

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